VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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A Project Phase-1 report on

AI-Enabled Smart Village Service Portal Using MERN Stack

Submitted in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering
in
Computer Science and Engineering

by

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ACHARYA INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belgaum
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Certificate

Certified that the Project Phase-1(BCS685) entitled **"AI-Enabled Digital Gram Panchayat Services Portal for Smart Rural Governance"** is a Bonafide work carried out by **Darshan M (1AY23CS065), Gajula Harshith (1AY23CS077), M Shankarlingeshwar (1AY23CS107), Shailaja(1AY23CS166)** in partial fulfillment for the award of degree of **Bachelor of Engineering in Computer Science & Engineering of Visvesvaraya Technological University**, Belagavi during the year **2025-2026.** It is certified that all corrections/ suggestions indicated for internal assessments have been incorporated in the Report deposited in the departmental library. This project phase-1 (BCS685) report has been approved as it satisfies the academic requirements with respect to project work prescribed for the **Bachelor of Engineering Degree**.

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DECLARATION

We the students of B.E. Computer Science & Engineering, Acharya Institute of Technology, Soladevanahalli, Bengaluru, hereby declare that the Project Phase 1 work entitled “AI-Enabled Digital Gram Panchayat Services Portal for Smart Rural Governance” is an authentic work carried out and successfully completed under the supervision and guidance of Mrs. Shrutika C Rampure, Assistant Professor, Department of Computer Science and Engineering, Acharya Institute of Technology, Bangalore. This dissertation work is submitted to Visvesvaraya Technological University in the partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering in Computer Science and Engineering department during the academic year 2025-26. We have not submitted the matter embodied to any other university or institution for the award of any other degree.

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ABSTRACT

The AI-Enabled Digital Gram Panchayat Services Portal for Smart Rural Governance is a web-based system designed to improve the accessibility and efficiency of rural administrative services. In traditional Gram Panchayat systems, villagers often face delays in accessing services such as certificates, complaint registration, bill payments, and information about government schemes due to manual processes and lack of centralized management. This project aims to create a digital platform where villagers can easily submit service requests, track application status, and receive updates online.

The portal also includes AI based features such as chatbot assistance, complaint classification, and voice input support to improve user interaction and reduce manual effort. The system is developed using the MERN Stack, which includes MongoDB for database management, Express.js and Node.js for backend processing, and React for frontend development.

AI functionalities are implemented using Python and Scikit-learn. The proposed system helps reduce paperwork, improve transparency, minimize processing delays, and provide better communication between villagers and Panchayat authorities. It supports digital transformation in rural governance by offering a smarter and more user friendly service platform.

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CHAPTER 1

INTRODUCTION

1.1. OVERVIEW:

The AI-Enabled Digital Gram Panchayat Services Portal for Smart Rural Governance is a web-based e-governance platform developed to improve the accessibility and efficiency of rural administrative services. In many villages, Gram Panchayat operations are still handled manually, which causes delays in service delivery, lack of transparency, and increased paperwork. Villagers often need to visit Panchayat offices multiple times for services such as certificate applications, complaint registration, bill payments, and information regarding government schemes.

The proposed system provides a centralized digital platform where citizens can access Panchayat services online. Users can register, log in, submit applications, upload documents, track request status, and receive notifications. The system also integrates AI-based features such as chatbot assistance, complaint classification, and voice input support to improve user interaction and reduce manual workload.

The portal is developed using the MERN Stack, which includes MongoDB, Express.js, React, and Node.js. AI functionalities are implemented using Python and Scikit-learn.

1.2. PROBLEM STATEMENT:

Traditional Gram Panchayat systems rely heavily on manual processes for handling citizen services. Villagers often experience delays in obtaining certificates, submitting complaints, and accessing government-related information. Lack of centralized digital management increases paperwork, reduces transparency, and creates communication gaps between villagers and Panchayat authorities.

Existing service systems mainly focus on basic application submission and do not provide intelligent support for complaint handling, automated assistance, or smart service management. Therefore, there is a need for a smart digital platform that improves rural governance and reduces manual administrative work.

1.3. MOTIVATION:

The motivation behind this project is to support digital transformation in rural governance and improve the quality of public service delivery in villages. Many rural citizens still face difficulties in accessing services because of manual verification, repeated office visits, and lack of technical support.

By integrating AI technologies such as chatbot assistance and complaint classification, the proposed system aims to simplify service access and provide faster responses. The project also focuses on reducing workload for Panchayat staff and improving transparency in administrative operations.

1.4. OBJECTIVES:

The main objectives of the project are:

- To develop a digital portal for Gram Panchayat services.
- To allow villagers to apply for services and track request status online.
- To implement AI-based features such as chatbot assistance and complaint classification.
- To reduce manual paperwork and processing delays.
- To improve communication between villagers and Panchayat staff.
- To provide a centralized and user-friendly rural governance platform.

1.5. SCOPE OF THE PROJECT:

The scope of the project includes providing digital access to rural administrative services through a web application. The system supports online complaint submission, certificate request management, service tracking, and AI-based user assistance.

The project is mainly designed as an academic prototype for smart rural governance and can be further expanded with additional features such as mobile applications, regional language support, GPS-based complaint tracking, and advanced analytics.

1.6. EXISTING SYSTEM:

In the existing system, most Gram Panchayat operations are performed manually. Citizens need to visit offices physically for submitting applications, tracking requests, and receiving approvals. Complaint handling and document verification are time-consuming processes and often lack transparency.

Although some online platforms provide basic services, they do not include intelligent automation, AI assistance, or smart complaint management features. Manual verification and lack of centralized data handling reduce operational efficiency.

1.7. PROPOSED SYSTEM:

The proposed AI-Enabled Digital Gram Panchayat Services Portal provides a centralized web-based platform for managing rural administrative services. Users can register, log in, submit requests, upload documents, track service status, and receive notifications digitally.

The system integrates AI technologies for chatbot support, automated complaint categorization, and voice-based interaction. The portal improves transparency, reduces manual workload, and enhances service delivery efficiency.

The proposed system also provides an admin dashboard for Panchayat staff to monitor applications, verify requests, manage complaints, and update service status effectively.

CHAPTER 2

LITERATURE REVIEW

2.1.E-GRAM PANCHAYAT:

The paper "E-Gram Panchayat" focuses on developing a digital platform for providing Panchayat services such as complaint registration, scheme information, certificate requests, and service tracking. The system improves transparency and reduces manual paperwork in village administration. The proposed model mainly supports web-based service access and administrative management for rural citizens. However, the system lacks AI-based automation features such as chatbot assistance, voice support, and smart complaint classification.

2.2. DIGITAL TRANSFORMATION AND LOCAL SELF-GOVERNANCE IN INDIA: EVALUATING THE IMPACT OF E-GOVERNANCE ON PANCHAYATI RAJ INSTITUTIONS:

This paper explains the role of e-governance in improving Panchayati Raj institutions in India. It discusses how digital systems help improve transparency, public participation, and service delivery in rural governance. The study also highlights challenges such as low digital literacy, infrastructure limitations, and lack of user-friendly platforms in villages. The paper suggests that intelligent automation and multilingual support can improve accessibility and efficiency in rural services.

2.3. E-GRAM PANCHAYAT:

The paper presents a web-based Panchayat management system that provides online services for certificates, complaints, and scheme management. It includes modules for citizens and administrators to manage requests and track service status. The proposed system improves digital communication between villagers and local authorities. However, the system does not include AI functionalities for automated assistance or predictive analysis, which limits smart decision-making capabilities.

2.4. GRAMCONNECT: A DIGITAL PLATFORM FOR GRAM PANCHAYAT SERVICES WITH CHATBOT SUPPORT:

This research introduces a digital Gram Panchayat platform integrated with chatbot support for citizen interaction. The system provides complaint registration, service access, alerts, and digital records management using modern web technologies. Chatbot integration improves user support and simplifies communication between villagers and Panchayat authorities. However, the system has limited AI analytics and lacks advanced automation features such as complaint prediction and intelligent service recommendations.

2.5. SMART DIGITAL APPLICATION FOR GRAM PANCHAYAT-DRIVEN RURAL WATER SUPPLY MANAGEMENT:

This paper focuses on digital monitoring of rural water supply systems using dashboards, complaint management, and maintenance tracking. The application helps authorities monitor water quality and pump operations effectively. The proposed system mainly addresses water supply management and does not provide a complete solution for all Panchayat services. Extending the system with AI-based issue prediction and multi-service integration can improve rural governance efficiency.

2.6. LITERATURE SURVEY SUMMARY TABLE:

Table 2.1 LITERATURE SURVEY

Sl. No	Tittle of the paper/journal	Publication details	Problem addressed in the paper	Authors approach	Shortfalls in the author's approach
1	E-Gram Panchayat	IEEE NKCon, 2023	Manual handling of Panchayat services	Developed web-based service and tracking portal	No AI-based assistance or analytics
2	Digital Transformation and Local Self-Governance in India	The Social Science Review, Vol 3, Issue 5, 2025	Lack of transparency and digital access in Panchayati Raj institutions	Implemented e-governance system for better governance	Infrastructure gaps and low digital literacy
3	Smart Digital Application for Gram Panchayat-Driven Rural Water Supply Management	IEEE ICISS, 2026	Rural water supply issue management	Created digital dashboard and complaint system	Focused only on water management services
4	GramConnect: A Digital Platform for Gram Panchayat Services with Chatbot Support	IEEE IC3ET, 2026	Difficulty in citizen support and communication	Integrated chatbot with complaint and record management	Limited AI prediction and automation
5	E-Gram Panchayat	AIJFR, Vol 7, Issue 1, 2026	Delay in rural administrative services	Developed web/mobile platform for complaints, certificates, and schemes	No AI features such as chatbot and smart automation

CHAPTER 3

REQUIREMENTS SPECIFICATION

3.1. FUNCTIONAL REQUIREMENTS:

Functional requirements describe the major services and operations provided by the proposed system.

- The system should allow users to register and log in securely.
- Users should be able to submit complaints digitally.
- The portal should support uploading complaint images and location details.
- The system should provide AI-based complaint classification.
- The platform should include AI chatbot assistance for user guidance.
- Users should be able to access village resource information such as hospitals, fair price shops, and health centers.
- The system should support Kannada voice input for complaint submission.
- The portal should provide a digital Panchayat notice board for announcements and Gram Sabha updates.
- Users should be able to track complaint and request status.
- The system should provide MGNREGA wage tracking information.
- The platform should include a health entitlement checker for government schemes.
- Admin should be able to monitor complaints and manage announcements.
- The system should send SMS or WhatsApp notifications to users.

3.2. NON-FUNCTIONAL REQUIREMENTS:

Non-functional requirements describe the performance and quality attributes of the system.

- The system should provide fast response time.
- The platform should support multiple users simultaneously.
- The portal should ensure secure storage of user data.
- The AI model should provide accurate complaint classification and chatbot responses.
- The user interface should be simple and easy to use.
- The system should support local language accessibility.
- The platform should provide reliable service without frequent downtime.
- The portal should maintain data consistency and integrity.

- Notification delivery should be efficient and timely.
- The system should be scalable for future enhancements.

3.3. SOFTWARE REQUIREMENTS:

The following software tools and technologies are used for developing the system.

- React for frontend development.
- Node.js for backend processing.
- Express.js for API creation and routing.
- MongoDB for database management.
- Python for AI implementation.
- Scikit-learn for machine learning operations.
- TensorFlow for advanced AI processing.
- OpenCV for image processing.
- Visual Studio Code for coding and debugging.
- GitHub for version control and project management.

3.4. HARDWARE REQUIREMENTS:

The following software tools and technologies are used for developing the system.

- Intel i5 / Ryzen 5 processor or above.
- Minimum 8 GB RAM.
- 256 GB SSD storage.
- CUDA-enabled GPU (optional for AI training).
- Stable internet connection.
- Desktop or laptop system for development and testing.

3.5. OTHER REQUIREMENTS:

Additional requirements needed for the system are:

- SMS/WhatsApp API for notifications and alerts.
- GPS or map integration for location-based complaint tracking.
- AI training dataset for chatbot and complaint classification.
- Cloud or internet connectivity for online services.
- Kannada language support tools for voice input.
- Development and testing tools for debugging and deployment.

CHAPTER 4

DESIGN

4.1. SYSTEM DESIGN:

System design describes the overall structure and working process of the proposed AI-Enabled Smart Rural Assistance and Panchayat Transparency Platform. The system is designed as a web-based application that connects villagers and Panchayat authorities through digital services and AI-based assistance.

The portal allows users to register, log in, submit complaints, access village resources, view announcements, and track requests. AI functionalities such as chatbot support, complaint classification, and Kannada voice input improve user interaction and reduce manual effort.

The system follows a client-server architecture where the frontend communicates with the backend APIs, and the backend interacts with the database and AI modules for processing user requests.

4.1.1. System architecture

The system architecture consists of the following components:

- Frontend developed using React.
- Backend server developed using Node.js and Express.js.
- Database management using MongoDB.
- AI modules implemented using Python and Scikit-learn.
- Notification services using SMS/WhatsApp APIs.
- GPS and location services for complaint tracking.

Working Flow:

- User interacts with frontend portal.
- Frontend sends request to backend API.
- Backend validates and processes request.
- AI module performs classification or chatbot processing.
- Data is stored/retrieved from database.
- Response is returned to user/admin dashboard.

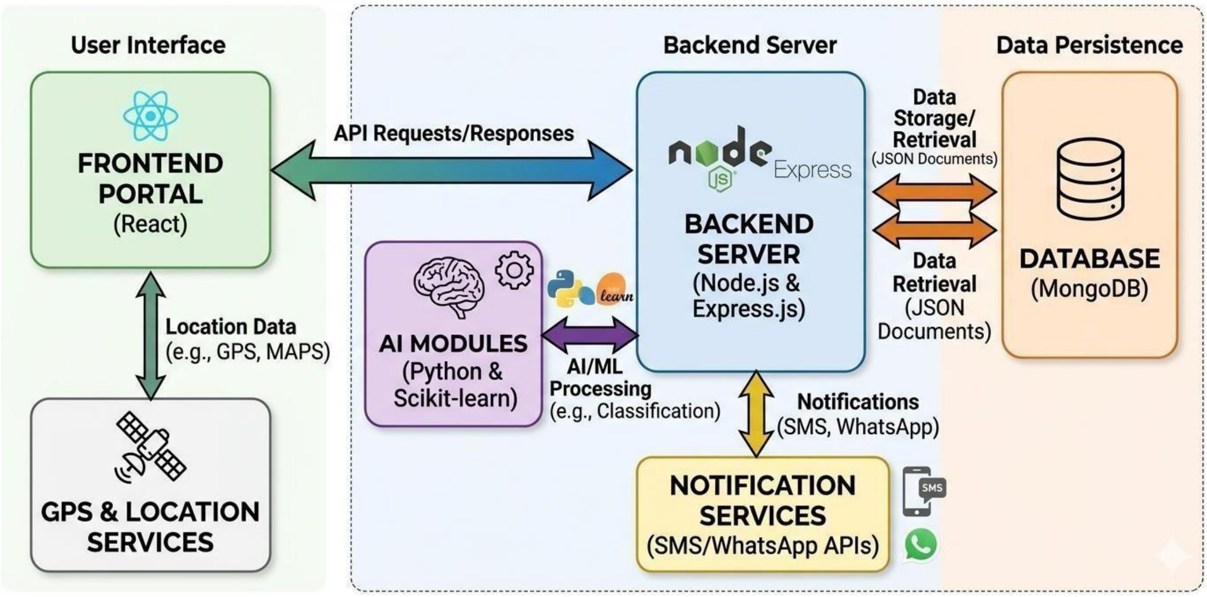

Fig 4.1.1.System architecture

4.1.2. Module design

The system is divided into multiple modules for efficient management and functionality.

User Management Module

- Handles user registration and login.
- Maintains authentication and profile management.

Complaint Management Module

- Allows users to submit complaints.
- Supports image upload and GPS location tracking.
- AI categorizes complaints automatically.

AI Chatbot Module

- Provides automated user support.
- Answers queries related to services and schemes.
- Supports local language interaction.

Village Resource Discovery Module

- Displays nearby hospitals, fair price shops, health centers, and community halls.
- Helps villagers access important local resources.

Digital Notice Board Module

- Displays Panchayat announcements and Gram Sabha notices.
- Sends updates to villagers digitally.

MGNREGA Wage Tracker Module

- Displays work history and wage details.
- Improves transparency for workers.

Health Entitlement Checker Module

- Suggests eligible government health schemes.
- Uses user details for recommendation.

Admin Dashboard Module

- Allows Panchayat staff to monitor complaints and services.
- Updates request status and manages announcements.

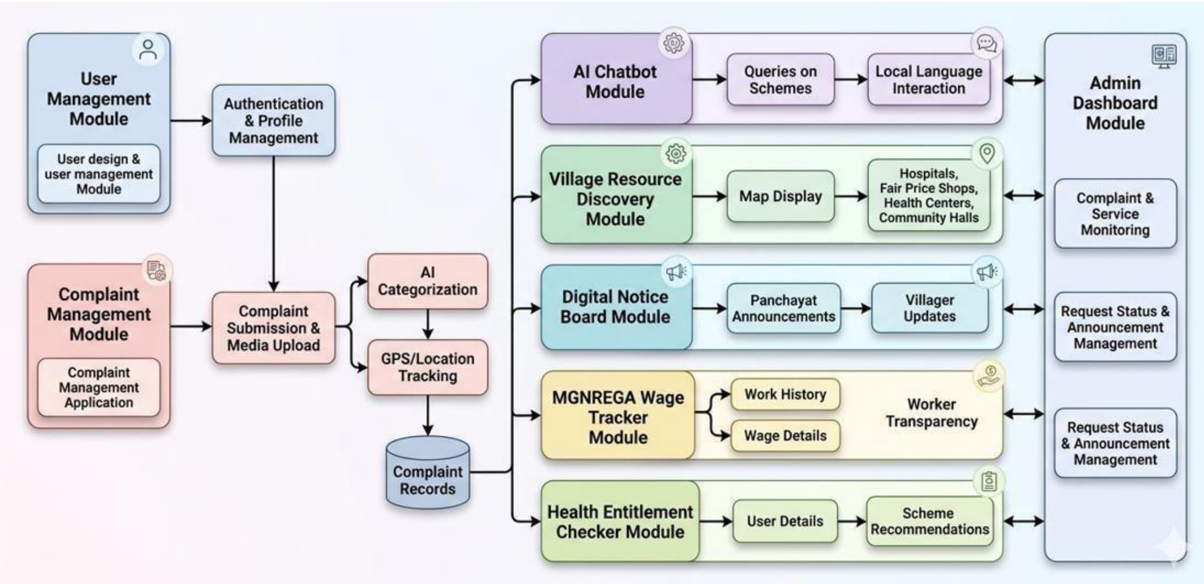

Fig 4.1.2 Module Design

4.2. DETAILED DESIGN:

The detailed design of the proposed AI-Enabled Smart Rural Assistance and Panchayat Transparency Platform explains the internal structure, workflow, and interaction between different modules of the system. The platform is designed to provide efficient communication between villagers and Panchayat authorities through digital services and AI-based assistance.

4.2.1. Class diagram

The class diagram represents the structure of the system and relationships between different classes.

Main classes used in the system include:

- User

- Admin
- Complaint
- Notification
- Resource
- Scheme
- WageTracker
- Chatbot

The User class interacts with complaints, schemes, and notifications. The Admin class manages complaints and announcements. AI-related classes handle chatbot responses and complaint categorization.

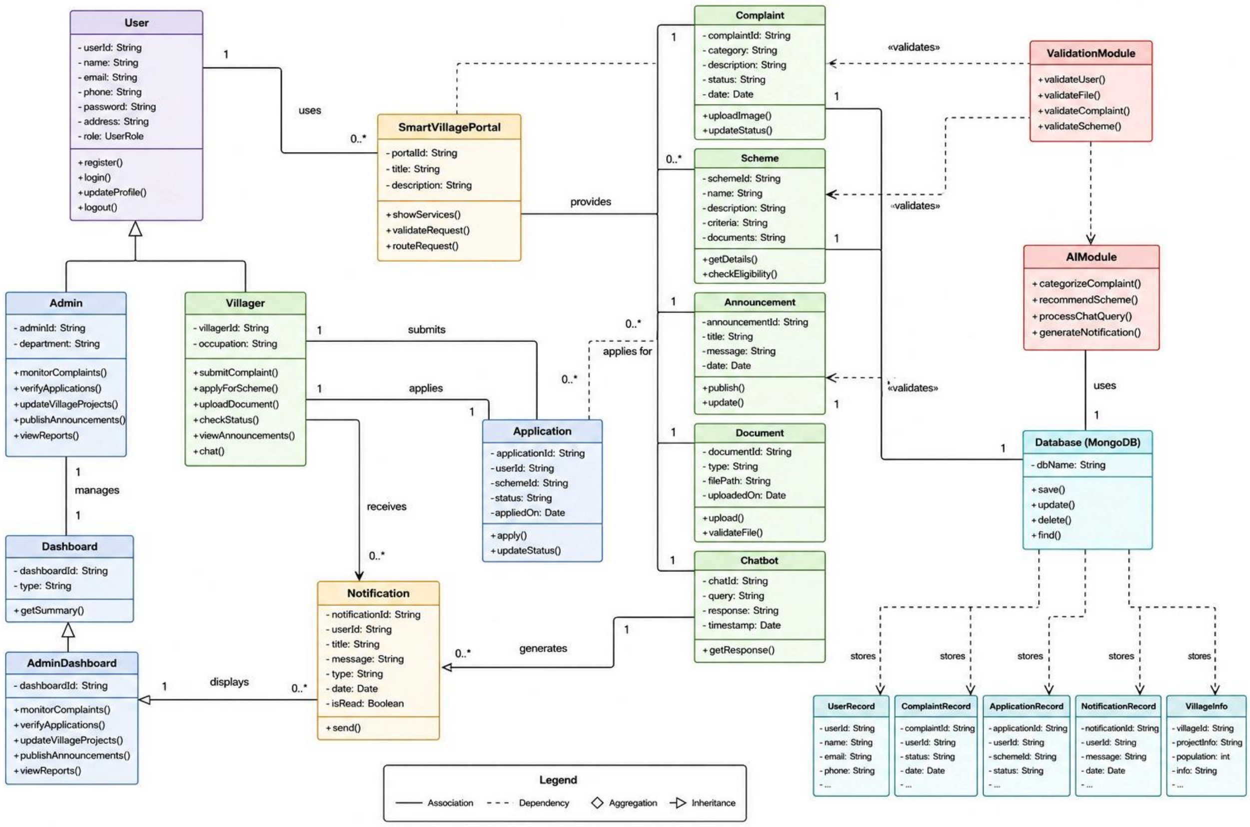

Fig 4.2.1 Class diagram

4.2.2. Activity diagram

The activity diagram represents the workflow of the system.

Main Activities:

- User registration and login.
- Complaint submission with image/location.
- AI complaint classification.

- Admin verification and status update.
- Notification delivery to users.
- Viewing village resources and announcements.

The activity flow helps understand the sequence of operations performed in the system.

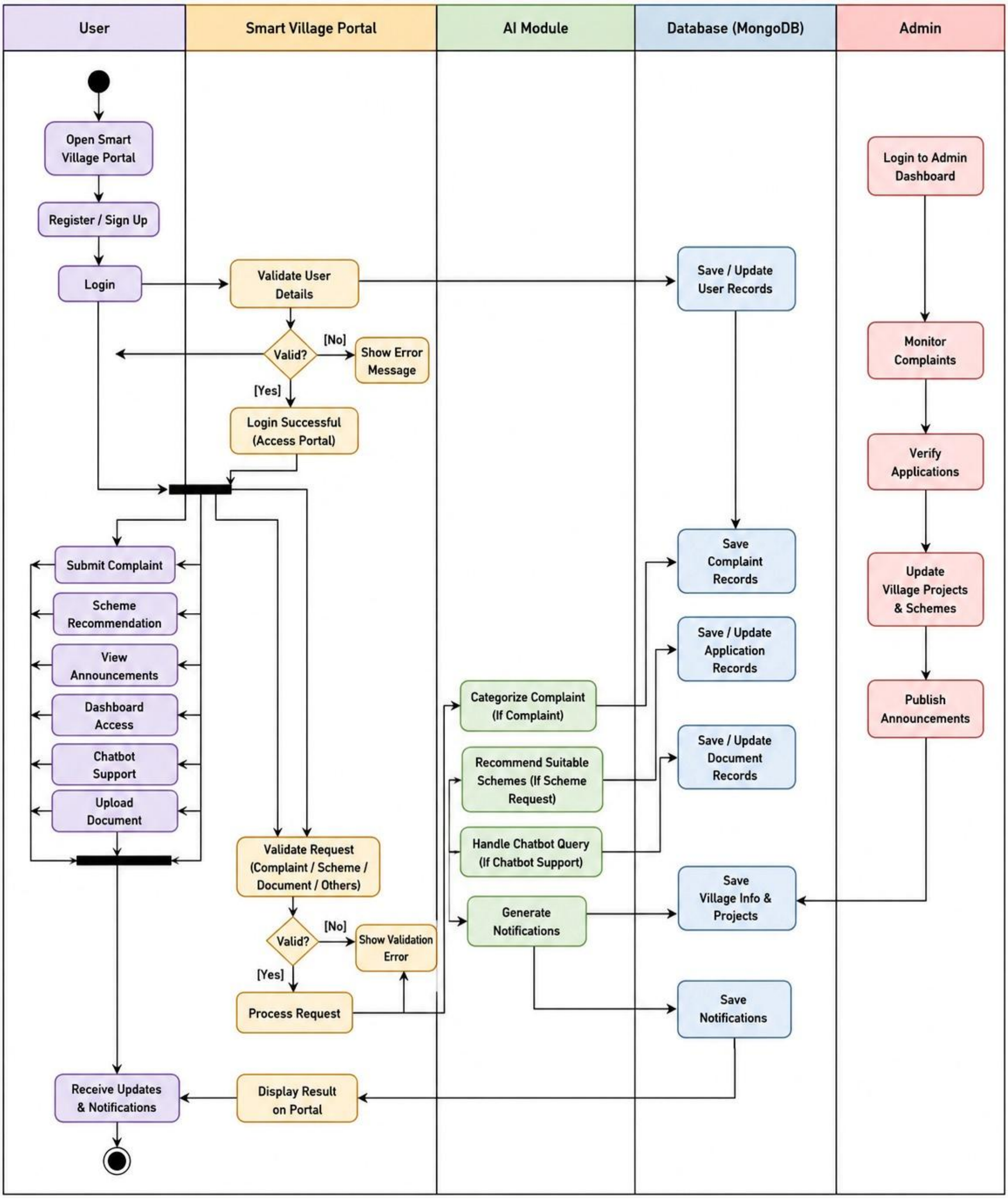

Fig 4.2.2 Activity Diagram

4.2.3. Data flow diagram

The data flow diagram describes how data moves through the system.

Data Flow:

- User submits request or complaint.
- Backend validates data.
- AI module processes complaint or chatbot query.
- Database stores user information and request details.
- Admin reviews and updates status.
- Notification is sent to user.

The DFD helps visualize data movement between users, system modules, and database.

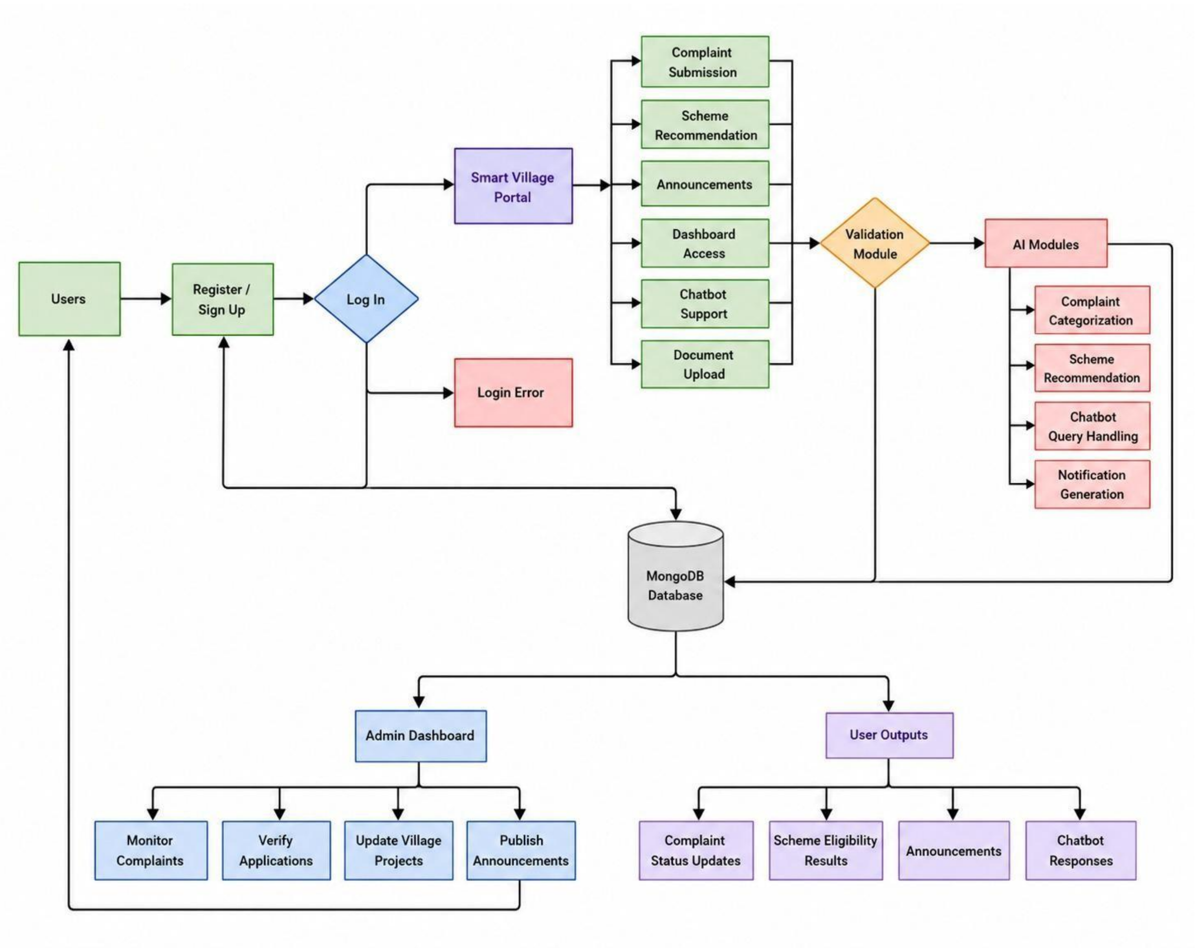

Fig 4.2.4. Data flow diagram

4.2.4. Scenarios

Scenario 1 – Complaint Registration

- User logs into portal.
- Uploads complaint image and location.
- AI categorizes complaint.
- Admin receives complaint notification.
- User tracks complaint status.

Scenario 2 – AI Chatbot Assistance

- User asks query in local language.
- Chatbot processes question.
- System provides automated response.

Scenario 3 – Digital Notice Board

- Admin posts village announcement.
- Villagers receive SMS/WhatsApp notification.
- Users view updates through portal.

Scenario 4 – Health Scheme Recommendation

- User enters eligibility details.
- System checks available schemes.

CHAPTER 5

PROJECT IMPLEMENTATION

5.1. MODULE-1: USER AUTHENTICATION MODULE

The User Authentication Module is responsible for user registration and login functionalities. It allows villagers and administrators to securely access the portal using valid credentials. User details are stored in MongoDB, and authentication is handled using backend APIs developed with Node.js and Express.js.

JavaScript

```javascript
const express = require('express');
const router = express.Router();
const User = require('./models/User');

router.post('/register', async (req, res) => {
   try {
      const user = new User({
         name: req.body.name,
         email: req.body.email,
         password: req.body.password
      });

      await user.save();
      res.status(200).json("User Registered Successfully");
   } catch (error) {
      res.status(500).json(error);
   }
});

module.exports = router;
```

The module uses Express.js routing to implement a registration API that handles user signup functionality. User details such as name, email, and password are received from the frontend application and processed by the backend server. These details are then stored securely in the MongoDB database using the User model. After successful registration, the system sends a success response back to the user confirming that the account has been created successfully.

5.2. MODULE-2: COMPLAINT MANAGEMENT MODULE

The Complaint Management Module allows villagers to submit complaints related to village issues such as water supply, road damage, sanitation, or electricity problems. The module stores complaint data in the database and helps Panchayat staff monitor and manage complaints efficiently.

```
const express = require('express');
const router = express.Router();
const Complaint = require('./models/Complaint');

router.post('/addComplaint', async (req, res) => {
   try {
      const complaint = new Complaint({
         userId: req.body.userId,
         complaintType: req.body.complaintType,
         description: req.body.description,
         status: "Pending"
      });

      await complaint.save();
      res.status(200).json("Complaint Submitted Successfully");
   } catch (error) {
      res.status(500).json(error);
   }
});

module.exports = router;
```

This module creates an API for complaint submission and management. Users can submit their complaint details through the frontend interface, and the submitted information is stored in the MongoDB database with a default status set to "Pending." The system allows Panchayat administrators to review the complaints and update their status through the admin dashboard, ensuring proper tracking and resolution of citizen issues.

CHAPTER 6

TESTING

6.1. TESTING:

Testing is an important phase in software development used to verify whether the system functions correctly according to the project requirements. It helps identify errors, improve performance, and ensure reliable operation of the application. In the proposed AI-Enabled Digital Gram Panchayat Services Portal, testing is performed to check the functionality of user registration, login, complaint submission, certificate request handling, AI chatbot responses, and database operations.

Different types of testing are carried out during project implementation to ensure system accuracy and efficiency. Functional testing is used to verify whether all modules work properly. User authentication testing ensures that only valid users can access the system. Database testing is performed to verify correct storage and retrieval of information from MongoDB. API testing checks communication between frontend and backend modules developed using Node.js and Express.js.

The AI features such as chatbot assistance and complaint classification are also tested to ensure accurate responses and proper categorization of complaints. Performance testing is conducted to verify that the system can handle multiple users and requests without significant delay.

The testing process ensures that the proposed system provides secure, accurate, and efficient digital services for rural governance.

CHAPTER 7

RESULTS AND DISCUSSION

7.1. RESULTS AND DISCUSSION:

The AI-Enabled Digital Gram Panchayat Services Portal was successfully developed and tested using the MERN Stack and AI technologies. The system provides a centralized platform for villagers to access rural administrative services digitally. Users can register, log in, submit complaints, apply for certificates, and track the status of their requests through a user-friendly web interface.

The implemented system successfully performs core functionalities such as user authentication, complaint management, request tracking, and admin monitoring. The backend APIs developed using Node.js and Express.js effectively communicate with the frontend developed using React. Data is securely stored and retrieved from MongoDB.

AI-based features such as chatbot assistance and complaint classification were integrated successfully to improve user interaction and reduce manual processing. The chatbot provides basic guidance related to Panchayat services, while complaint classification helps categorize user issues efficiently.

The proposed system reduces paperwork, minimizes delays in service handling, and improves transparency between villagers and Panchayat authorities. The portal also improves accessibility by allowing users to access services online without repeatedly visiting Panchayat offices.

Overall, the project demonstrates how AI and modern web technologies can support smart rural governance and improve the efficiency of public service delivery in villages.

CHAPTER 8

CONCLUSION & FUTURE SCOPE

8.1. CONCLUSION:

The AI-Enabled Digital Gram Panchayat Services Portal for Smart Rural Governance was developed to improve the efficiency and accessibility of rural administrative services. The proposed system provides a centralized digital platform where villagers can access services such as complaint submission, certificate request management, and application status tracking through an online portal

The system successfully integrates modern web technologies and AI-based features to reduce manual workload and improve transparency in rural governance. The implementation using the MERN Stack — MongoDB, Express.js, React, and Node.js — provides an efficient and user-friendly platform. AI functionalities developed using Python and Scikit-learn help automate complaint classification and chatbot support.

The project demonstrates how digital technologies can support smart village development by improving service delivery, reducing delays, and enhancing communication between villagers and Panchayat authorities.

8.2. FUTURE SCOPE:

The proposed system can be further enhanced with advanced technologies and additional features to improve rural governance services. Future improvements may include developing a mobile application for easier accessibility, integrating regional language support for villagers, and implementing advanced AI models for better chatbot responses and predictive analysis.

The system can also be extended with GPS-based complaint tracking, image-based issue detection, and automated notification systems. Features such as voice-enabled interaction, online document verification, and cloud-based data management can improve usability and scalability.

In the future, the project can be integrated with smart village initiatives to support intelligent decision-making, real-time monitoring, and efficient management of rural administrative services.

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